

Appendix

Laplace Transform

It is one of the great operational tools of applied mathematics: it turns calculus into algebra.

Definition: For a function $f(t)$ defined for $t \geq 0$, the Laplace Transform is

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

where s is a complex variable, $s = \sigma + i\omega$, provided the improper integral converges.

The integral converges when f is of exponential order: $|f(t)| \leq Me^{ct}$ for some M, c and t large. Then $F(s)$ exists for $\Re(s) > c$.

This tool is widely used in engineering and physics to analyze control systems, circuits, and differential equations (Campbell and Haberman 247-250). The idea: differentiation in t -domain becomes multiplication by s in s -domain.

Historical Note

While named after Pierre-Simon Laplace who used it in probability in 1782, the essential ideas appeared much earlier in Leonhard Euler's work from the 1730s and 1750s (Deakin 264-267). Euler used similar integral transforms $\int X(x)e^{ax}dx$ to solve differential equations decades before Laplace formalized the method, demonstrating how mathematical concepts often exist in practice before receiving formal names and notation (Deakin 268-269).

The transform remained relatively obscure until Oliver Heaviside rediscovered and popularized operational methods $p = d/dt$ in the 1880s-1890s for solving electrical circuit problems (Widder 419-420). Heaviside could solve telegraph equations "by inspection" but without rigorous justification. Thomas Bromwich, Gustav Doetsch and David Widder later made it rigorous via complex inversion.

How to Compute and Invert

Forward: Evaluate the improper integral, often using integration by parts, or using tables for common functions (MIT OCW).

Example: $\mathcal{L}\{1\} = \int_0^{\infty} e^{-st} dt = 1/s, \Re(s) > 0$.

Inverse: Recover $f(t)$ from $F(s)$, typically using partial fraction decomposition and inverse transform tables (Ungar 786-788; Widder 179-180). The rigorous inversion is Bromwich integral:

$$f(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} e^{st} F(s) ds$$

In practice, however, the inversion process can be remarkably intuitive once patterns are recognized, allowing practitioners to work "by inspection" without formal contour integration (Ungar 789-791). Most engineers use tables.

Beyond basic forms, the Laplace transform has been computed for remarkably complex functions including Bessel functions $J_n(t)$ (Spiegel 329-330), error functions $\text{erf}(t)$ (Opatowski 392), and the psi (digamma) function $\psi(t)$ (Dixit 593-600). These specialized results connect the Laplace transform to deep areas of analysis including the gamma function and Euler's constant

(Pribitkin 241-245). Generalizations extend the classical Laplace transform to time scales and conformable derivatives, broadening applicability to discrete-continuous hybrid systems (Thang et al. 1699-1705).

Properties — Why It Works

Let $F(s) = \mathcal{L}\{f\}$, $G(s) = \mathcal{L}\{g\}$ (Campbell and Haberman 251-255; Guggenheimer 196-198):

1. Linearity:

$$\mathcal{L}\{af + bg\} = aF + bG$$

Transform respects superposition.

2. Differentiation in t becomes multiplication in s :

$$\mathcal{L}\{f'\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$$

$$\mathcal{L}\{f^{(n)}\} = s^nF(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$$

This is the core advantage: ODE becomes algebraic. Initial conditions $f(0), f'(0)$ are automatically incorporated — ideal for transient analysis (Guggenheimer 199-200).

3. Differentiation in s :

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$$

4. Integration:

$$\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \frac{F(s)}{s}$$

5. Time shift:

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

where $u(t-a)$ is unit step.

6. Frequency shift / Damping:

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

7. Convolution — Transforms multiply:

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$$

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

This is crucial for LTI systems: output = input * impulse response, so in s -domain $Y(s) = H(s)X(s)$.

8. Initial and Final Value Theorems:

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s)$$

$$f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

if limits exist.

Advantages of Laplace Transform

- **Simplification:** Converts complicated linear integro-differential equations with constant coefficients into easy algebraic equations (Lunardi 185-188). Solve algebraically, then invert.
- **Initial Conditions Built-In:** Unlike Fourier, automatically incorporates $f(0), f'(0)$ – making it ideal for initial value problems, transient circuit analysis, mechanical vibrations.
- **System Function / Transfer Function:** For ODE $a_n y^{(n)} + \dots + a_0 y = b_m x^{(m)} + \dots$, $H(s) = Y(s)/X(s) = \frac{b_m s^m + \dots}{a_n s^n + \dots}$. Poles of $H(s)$ give stability, frequency response, without solving entire equation (Campbell and Haberman 258-262).
- **Matrix Exponentials:** Provides elegant methods for computing e^{At} in systems $\mathbf{x}' = A\mathbf{x}$ using $\mathcal{L}\{e^{At}\} = (sI - A)^{-1}$ and algorithmic partial fractions (Adkins and Davidson 267-273).
- **Handles Discontinuities:** Easily handles Heaviside steps, Dirac deltas, periodic forcing – where classical methods break.

Table of Common Transforms

$f(t), t \geq 0$	$F(s) = \mathcal{L}\{f\}$	ROC	Notes
1 (unit step)	$\frac{1}{s}$	$\Re(s) > 0$	$t^n, n \in \mathbb{N}_0$ $\frac{n!}{s^{n+1}}$ (Pribitkin 238-240) $\Re(s) > 0$ extends to $\Gamma(n+1)/s^{n+1}$ for non-integer
e^{at}	$\frac{1}{s-a}$	$\Re(s) > a$	$\sin(bt)$ $\frac{b}{s^2+b^2}$ (Efthimiou 376-378) $\Re(s) > 0$ $\cos(bt)$ $\frac{s}{s^2+b^2}$ $\Re(s) > 0$ $\sinh(bt)$ $\frac{b}{s^2-b^2}$ $\Re(s) > b$ $\cosh(bt)$ $\frac{s}{s^2-b^2}$ $\Re(s) > b$ $e^{at} \sin(bt)$ $\frac{b}{(s-a)^2+b^2}$ $\Re(s) > a$ damping shifts $e^{at} \cos(bt)$ $\frac{s-a}{(s-a)^2+b^2}$ $\Re(s) > a$ $t^n e^{at}$ $\frac{n!}{(s-a)^{n+1}}$ $\Re(s) > a$ $\delta(t)$ Dirac 1 all s $u(t-a)$ step at a $\frac{e^{-as}}{s}$ $\Re(s) > 0$ Bessel $J_0(at)$ $\frac{1}{\sqrt{s^2+a^2}}$ (Spiegel) $\Re(s) > 0$

In short: Laplace transform trades a hard calculus problem for an easier algebra problem in the s -domain, solves it, and trades back. Its power comes from linearity, the differentiation rule, and convolution theorem – which together make it the natural language of linear time-invariant systems.

Taylor Series

For a function $f(x)$ that is infinitely differentiable near a , the Taylor series is the infinite polynomial that matches f and all its derivatives at a :

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

When $a = 0$, this special case is called the Maclaurin series. The remarkable fact is that every power series with positive radius of convergence is actually a Taylor series of its sum function (Meyerson 51-52), creating a fundamental equivalence between locally convergent power series and analytic functions.

The Core Idea: Local Polynomial Approximation

The Taylor series represents one of mathematics' most powerful ideas: any sufficiently smooth curve can be approximated locally by polynomials (Eves 40-45).

Think iteratively:

- **0th order:** $P_0(x) = f(a)$ – constant approximation. Matches value.
- **1st order:** $P_1(x) = f(a) + f'(a)(x - a)$ – tangent line. Matches value and slope. This is linear approximation.
- **2nd order:** $P_2(x) = P_1 + \frac{f''(a)}{2}(x - a)^2$ – osculating parabola. Matches curvature too.
- **nth order:** P_n matches $f, f', \dots, f^{(n)}$ at a .

Each term corrects the error of previous polynomial to higher order: $f(x) - P_n(x) = O((x - a)^{n+1})$.

This is why Taylor series are ubiquitous in physics: for small $(x - a)$, low-order Taylor polynomial is excellent. $\sin x \approx x$, $e^x \approx 1 + x$, $\sqrt{1 + x} \approx 1 + x/2$ are all first-order Taylor.

Remainder and Convergence

Writing infinite series is formal; does it equal $f(x)$?

Taylor's Theorem with Remainder: If f is $(n + 1)$ -times differentiable, then for x near a ,

$$f(x) = P_n(x) + R_n(x)$$

where P_n is n th Taylor polynomial, and R_n can be expressed in several forms. Lagrange form:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - a)^{n+1}$$

for some ξ between a and x . This is a direct generalization of the Mean Value Theorem, which is the $n = 0$ case (Spiegel 263-266). Other forms: Cauchy, integral form:

$$R_n(x) = \frac{1}{n!} \int_a^x (x - t)^n f^{(n+1)}(t) dt$$

This connection guarantees that error in truncating after n terms can be bounded and estimated if you can bound $f^{(n+1)}$.

Convergence: The series converges to $f(x)$ iff $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$.

Taylor series converge to the original function within a specific radius of convergence R , but can diverge outside this region or at pathological points (Erdős et al. 262-266).

- If f is analytic (complex differentiable) near a , then R is distance from a to nearest singularity in complex plane.
Example: $f(x) = 1/(1 + x^2)$ has Maclaurin series $1 - x^2 + x^4 - \dots$ with $R = 1$, because singularities at $\pm i$ are distance 1 from 0, even though f is smooth on $\mathbb{R}$.
- If f is C^∞ but not analytic, series may converge to wrong function. Classic counterexample: $f(x) = e^{-1/x^2}$ for $x \neq 0$, $f(0) = 0$. All derivatives at 0 are 0, so Maclaurin series is 0, which does not equal $f(x)$ for $x \neq 0$. So infinite differentiability is not enough.
- Coefficients' behavior determines convergence: bounded coefficients $|f^{(n)}(a)/n!| \leq M$ ensures convergence at least within unit circle $|x - a| < 1$ (Duffin and Schaeffer 141-145). More generally, $R = 1/\limsup |c_n|^{1/n}$ by Cauchy-Hadamard.

This enables both theoretical analysis and practical computation (Widder 126-130). Integration methods like Romberg integration leverage Taylor series to achieve high-precision numerical results by extrapolating trapezoidal errors which have even-power Taylor expansion (Rozema 284-288).

Standard Maclaurin Series — The Toolbox

These converge for stated x and are used constantly:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad |x| < \infty$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \quad |x| < \infty$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \quad |x| < \infty$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots \quad |x| < 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad |x| < 1, x = 1 \text{ converges conditionally}$$

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \cdots \quad |x| < 1$$

These generate others by substitution, differentiation, integration.

Operations with Taylor Series

If $f = \sum c_n(x-a)^n$ and $g = \sum d_n(x-a)^n$ converge for $|x-a| < R$:

- Addition, multiplication by polynomial multiplication (Cauchy product)
- Composition if inner series has zero constant term
- Differentiation and integration term-by-term inside R

This is why power series are so powerful: calculus on functions becomes algebra on coefficients.

Deep connection: The deep connection between Taylor series and Laplace transforms provides powerful tools for solving differential equations (Euler 305-307). Formally, $\mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t) dt$ can be expanded by expanding f in Taylor series, and Laplace transform of t^n is $n!/s^{n+1}$, linking Taylor coefficients to asymptotic expansion of $F(s)$ at infinity.

Why Taylor Series Matter

1. **Local linearization** is foundation of calculus, Newton's method, stability analysis.
2. **Definition of functions:** Many functions (e^x , $\sin x$, $\cos x$) are *defined* in analysis by their Taylor series, proving existence.
3. **Numerical computation:** Computers evaluate transcendental functions by truncated Taylor / Chebyshev polynomials.
4. **Physics:** Small oscillations, perturbation theory, relativity corrections — all Taylor expansions: $(1 - v^2/c^2)^{-1/2} = 1 + v^2/2c^2 + \cdots$.
5. **Analytic continuation and complex analysis:** Taylor series is the seed from which analytic functions grow.

In one line: Taylor series says that if you know a function and all its derivatives at one point, you know the function everywhere within its radius of convergence — local information determines global behavior for analytic functions.

Eigenvalue / Eigenvector

Formally, given a linear transformation represented by matrix $A \in \mathbb{F}^{n \times n}$, a nonzero vector v is an eigenvector with eigenvalue $\lambda \in \mathbb{F}$ if

$$Av = \lambda v$$

The transformation takes vector v and multiplies it by a scalar factor λ , preserving its direction up to flip: if $\lambda > 0$ direction preserved, $\lambda < 0$ reversed, $|\lambda|$ stretches, $|\lambda| < 1$ compresses. Zero vector is excluded by definition because $A0 = \lambda 0$ holds for all λ trivially (Chu 1-5).

The eigenvalue problem is one of the most fundamental in all of mathematics, appearing in differential equations, quantum mechanics, structural engineering, graph theory, and data analysis (Chu 5-10). It is the nonlinear heart of linear algebra.

Why $Av = \lambda v$ Matters — Geometric Intuition

Most vectors are rotated and stretched by A . Eigenvectors are the special directions where A acts purely as scaling.

- For rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ in $\mathbb{R}^2$, no real eigenvector — every nonzero vector is rotated 90° . Complex eigenvalues $\pm i$ appear.
- For shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, one eigenvector direction $e_1 = (1, 0)$, eigenvalue 1 — horizontal lines stay horizontal.
- For diagonal $D = \text{diag}(2, 3)$, e_1 stretched by 2, e_2 by 3 — axes are eigenvectors.
- If A describes deformation of an elastic body, eigenvectors are principal directions of strain.

If you have a basis of eigenvectors, A becomes diagonal in that basis — the transformation decouples into independent 1D scalings. That is the goal of diagonalization.

Finding Eigenvalues — Characteristic Equation

Finding eigenvalues requires solving the characteristic equation (Tisseur and Meerbergen 235-240).

$Av = \lambda v \iff (A - \lambda I)v = 0$ has nonzero solution v iff $A - \lambda I$ is singular:

$$\det(A - \lambda I) = 0$$

This is a degree n polynomial in λ — the characteristic polynomial $p_A(\lambda)$.

For $n \times n$ matrix, this yields n eigenvalues counting algebraic multiplicity over $\mathbb{C}$ by Fundamental Theorem of Algebra, though they may be complex even when A is real. Example: rotation above gives $p(\lambda) = \lambda^2 + 1$, roots $\pm i$.

Steps:

1. Compute $p_A(\lambda) = \det(A - \lambda I)$.
2. Find roots λ_i — eigenvalues.
3. For each λ_i , solve $(A - \lambda_i I)v = 0$ to find eigenspace $E_{\lambda_i} = \ker(A - \lambda_i I)$. Dimension is geometric multiplicity $\leq$ algebraic multiplicity.

If algebraic = geometric for all λ , A is diagonalizable: $A = PDP^{-1}$ where $D = \text{diag}(\lambda_i)$ and columns of P are eigenvectors. Then $A^k = PD^kP^{-1}$ easy, and ODE $x' = Ax$ solved by $x(t) = \sum c_i e^{\lambda_i t} v_i$.

If not diagonalizable, Jordan normal form $A = PJP^{-1}$ with Jordan blocks.

Numerical computation does NOT use $\det(A - \lambda I)$ for $n > 4$ — polynomial root-finding is unstable. Instead QR iteration, power iteration, Lanczos/Arnoldi for large sparse.

Spectrum of Generalizations

The standard problem $Av = \lambda v$ generalizes:

1. Generalized eigenvalue problem:

$$Av = \lambda Bv$$

Appears in vibration: $Kx = \lambda Mx$ where K stiffness, M mass. If B invertible, equivalent to $B^{-1}Av = \lambda v$, but better solved without inversion.

2. Quadratic eigenvalue problem (QEP):

$$(\lambda^2 M + \lambda C + K)x = 0$$

Arises in vibration analysis, acoustic modeling, and fluid-structure interaction, generalizing standard problem to account for mass M , damping C , and stiffness K (Tisseur and Meerbergen 240-250). Second-order system $M\ddot{x} + C\dot{x} + Kx = 0$ with ansatz $x = e^{\lambda t}v$ yields QEP. Linearized to $2n \times 2n$ problem $\begin{pmatrix} 0 & I \\ -K & -C \end{pmatrix}$.

3. Polynomial and nonlinear eigenvalue problems:

$$P(\lambda)x = \left(\sum_{k=0}^m \lambda^k A_k\right)x = 0$$

$T(\lambda)x = 0$ nonlinear in λ . Arise in delay equations, photonics.

4. Singular value decomposition relation: For A non-square, $A^T Av = \sigma^2 v$ gives singular values σ .

Inverse and Structured Problems

Inverse eigenvalue problem: Construct matrix with specified eigenvalues — given $\{\lambda_i\}$, find A with properties (symmetric, nonnegative, banded) having those eigenvalues. Has applications in control theory, system identification, molecular structure determination, and designing mass-spring systems with given frequencies (Chu 10-20). Highly nontrivial — not all spectra realizable.

Structured eigenvalue problems: If A symmetric/Hermitian: eigenvalues real, eigenvectors orthogonal, always diagonalizable. This is spectral theorem — basis of quantum mechanics where observables are Hermitian, eigenvalues are measurement outcomes. If A orthogonal/unitary: eigenvalues $|\lambda| = 1$. If A stochastic: $\lambda = 1$ is largest (Perron-Frobenius).

Random matrix theory: When entries of A random, eigenvalue distributions exhibit phase transitions and universal behavior — Wigner semicircle for symmetric Gaussian, circular law for non-Hermitian, Tracy-Widom for largest eigenvalue — with applications to statistics (PCA), nuclear physics (energy levels), wireless communications, and finance (Baik et al. 1643-1650).

Why Eigenvalues Dominate Applications

- 1. Differential equations:** $x' = Ax$ solution $x(t) = e^{At}x_0$. Growth/decay determined by $\Re(\lambda)$. $\Re(\lambda) < 0$ stable.
- 2. Stability:** Linearized dynamical system stable iff all eigenvalues in left half-plane (continuous) or unit disk (discrete).

3. **Vibrations:** Natural frequencies $\omega = \sqrt{\lambda}$ of $Kx = \lambda Mx$. Resonance.
4. **Quantum mechanics:** Schrödinger $H\psi = E\psi$ – eigenvalues E are energy levels.
5. **Data analysis:** Covariance matrix C eigenvalues are variances along principal components – PCA. Largest eigenvectors = directions of maximal variation. PageRank eigenvalue 1 of Google matrix.
6. **Graph theory:** Adjacency matrix eigenvalues encode connectivity, expansion, clustering.

Examples in Physics and Engineering

Musical Instruments (Resonance) – Eigenvalues You Can Hear

Musicians intuitively understand that instruments have "sweet spots" where certain notes resonate – they are finding physical eigenmodes without solving differential equations (Tisseur and Meerbergen 270-275). Every note an instrument plays is an eigenvalue; every timbre is a superposition of eigenfunction coefficients.

The Model: Ideal String Fixed at Both Ends

A guitar string fixed at both ends, length L , under tension T , with mass per length μ , obeys the 1D wave equation derived from $F = ma$ on a small element:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

where $u(x, t)$ is transverse displacement at position x and time t , and

$$c = \sqrt{\frac{T}{\mu}}$$

is wave speed. Higher tension = faster wave = higher pitch; heavier string = slower wave = lower pitch.

Boundary conditions: $u(0, t) = u(L, t) = 0$ – fixed at nut and saddle, cannot move. Initial conditions: shape $u(x, 0) = f(x)$ and velocity $u_t(x, 0) = g(x)$ from pluck.

Separation of Variables → Eigenvalue Problem

Seek standing wave solutions where spatial and temporal parts separate: $u(x, t) = X(x) \cdot \cos(\omega t)$ or $\sin(\omega t)$. Plug into wave equation:

$$\frac{X''}{X} = -\frac{\omega^2}{c^2} = -k^2 = \text{constant}$$

This yields the spatial eigenvalue problem – a Helmholtz equation in 1D:

$$\frac{d^2 X}{dx^2} = -k^2 X, \quad X(0) = X(L) = 0, \quad \text{where } k = \frac{\omega}{c}$$

Here d^2/dx^2 is the linear operator, X is eigenfunction, $-k^2$ is eigenvalue. Boundary conditions quantize k .

General solution: $X(x) = A \sin(kx) + B \cos(kx)$. $X(0) = 0 \implies B = 0$. $X(L) = 0 \implies A \sin(kL) = 0$. For nontrivial $A \neq 0$, need $\sin(kL) = 0$:

$$k_n L = n\pi, \quad n = 1, 2, 3, \dots$$

Only discrete k allowed – quantization emerges from boundary conditions. This is exactly like particle in a box in quantum mechanics.

Eigenfunctions and Eigenvalues

Eigenfunctions (mode shapes):

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \dots$$

These are the only shapes the string can vibrate in without "twisting into chaos" — each is an eigenfunction of d^2/dx^2 with eigenvalue $-k_n^2 = -(n\pi/L)^2$. They are orthogonal: $\int_0^L X_n X_m dx = 0$ for $n \neq m$. They form basis for any shape — Fourier sine series.

- $n = 1$: one hump, no interior nodes. Whole string moves in phase.
- $n = 2$: two humps, one node in middle at $x = L/2$ that stays still.
- $n = 3$: three humps, two nodes at $L/3, 2L/3$.
- n : n humps, $n - 1$ interior nodes.

Eigenvalues (frequencies):

$$\omega_n = ck_n = \frac{n\pi c}{L}, \quad f_n = \frac{\omega_n}{2\pi} = \frac{nc}{2L} = \frac{n}{2L} \sqrt{\frac{T}{\mu}}$$

Frequencies are integer multiples of fundamental — harmonic series. This harmonicity is why string sounds musical. Inharmonic instruments like drums have f_n not integer multiples.

For a guitar A string: $L \approx 0.65$ m, $f_1 = 110$ Hz (A2). Then $c = 2Lf_1 = 143$ m/s, implies $T = \mu c^2$. With typical $\mu \approx 0.005$ kg/m, $T \approx 102$ N.

- $n = 1$: Fundamental, 110 Hz — dominant pitch you hear, amplitude largest.
- $n = 2$: First overtone / second harmonic, 220 Hz — octave higher (A3).
- $n = 3$: Second overtone, 330 Hz — octave+fifth (E4 approx).
- $n = 4$: Third overtone, 440 Hz — two octaves (A4).
- $n = 5$: 550 Hz — etc., diminishing amplitude.

Superposition — What Happens When You Pluck

When you pluck, initial displacement $f(x)$ is triangular — pulled at point x_0 . The string "knows" its eigenmodes instinctively — physics forces vibration into these patterns. Any initial shape decomposes into sum of eigenfunctions (Fourier's insight):

$$u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right) \cos(2\pi f_n t + \phi_n)$$

Coefficients a_n depend on *where* you pluck, given by projection:

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

If pluck at center $x_0 = L/2$:

$$a_n \propto \frac{1}{n^2} \sin\left(\frac{n\pi}{2}\right)$$

So $a_2, a_4, a_6 = 0$ — even harmonics have node at center, cannot be excited by plucking at node. Result: hollow, clarinet-like, only odd harmonics.

If pluck near bridge $x_0 \approx 0.1L$: $\sin(n\pi x_0/L)$ non-zero for many n , $a_n \sim 1/n$ slowly decaying — many high harmonics excited, bright, twangy timbre (Tisseur and Meerbergen 250-255). Classical guitarists move plucking hand to control timbre — sul tasto (near fingerboard) = soft, few harmonics; ponticello (near bridge) = bright, many harmonics.

Damping C in real string makes higher modes decay faster: $u_n \sim e^{-\zeta_n t} \cos(\omega_{d,n} t)$, so brightness fades.

Every Instrument is an Eigenvalue Problem

- **Violin, piano, harp:** Same as guitar, but stiffness adds term $EIu_{xxxx} \rightarrow$ slight inharmonicity $f_n \approx n f_1(1 + Bn^2)$, why piano octaves are stretched when tuned.
- **Flute / organ pipe (air column):** Pressure p obeys wave equation. Open-open pipe: $p = 0$ at both ends $\rightarrow$ same eigenvalues $f_n = nc/2L$. Closed-open (clarinet): node at closed, antinode at open $\rightarrow f_n = (2n - 1)c/4L$ — only odd harmonics, hence clarinet sounds different from flute (Chu 25-30).
- **Drum membrane (2D):** $\nabla^2 X = -k^2 X$ on disk with $X = 0$ at rim. Eigenfunctions are Bessel functions $J_m(k_{mn}r) \cos(m\theta)$, eigenvalues are zeros of Bessel J_m . Frequencies not harmonic — drum sounds inharmonic.
- **Room acoustics:** Same wave equation in 3D box — eigenvalues are room modes causing boomy bass at $f_{n_x, n_y, n_z} = (c/2) \sqrt{(n_x/L_x)^2 + (n_y/L_y)^2 + (n_z/L_z)^2}$.

Google's Original PageRank Algorithm — The \$25 Billion Eigenvector

When you search Google and trust that the top result is probably most relevant, you're relying on eigenvector mathematics you've never seen (Bryan and Leise 580-581). The formalism $Av = \lambda v$ and characteristic polynomials $\det(A - \lambda I) = 0$ provide precision, but the conceptual competence — recognizing special directions, resonant patterns, influential positions, and dominant modes — operates continuously in perception, music, social navigation, and spatial reasoning (Schonefeld 318-319).

The \$25 Billion Eigenvector: In 2006, Google's market value was tied to this eigenvector computation running on billions of pages (Bryan and Leise 569). Larry Page and Sergey Brin's 1998 Stanford paper's brilliance: reducing the subjective problem of "importance" to an objective eigenvalue problem (Langville and Meyer 135-145). Modern search uses hundreds of factors, but PageRank's eigenvector remains foundational as a centrality signal (Langville and Meyer 145-155).

The Core Idea: Importance as Recursive Voting

Old search counted keywords. Page and Brin asked: what if a page's importance is determined by who links to it?

Idea: A link $j \rightarrow i$ is a vote by j for i . But not all votes equal — a vote from an important page counts more than a vote from an obscure page. So:

$$\text{importance}(i) = \sum_{j \text{ links to } i} \frac{\text{importance}(j)}{\text{outlinks}(j)}$$

This is recursive: importance of i depends on importance of j , which depends on who links to j , etc. This is exactly $\pi = H\pi$ — an eigenvector problem.

Imagine simplified web with 4 pages. Define hyperlink matrix H where

$$H_{ij} = \frac{1}{n_j} \text{ if page } j \text{ links to page } i, \quad 0 \text{ otherwise}$$

n_j = number of outlinks from page j . Column j sums to 1 if j has outlinks — H is column-stochastic — representing how j distributes its importance equally among pages it links to.

Suppose:

- Page 1 links to pages 2,3,4 ($n_1 = 3$)
- Page 2 links to page 1 ($n_2 = 1$)
- Page 3 links to pages 1,4 ($n_3 = 2$)
- Page 4 links to pages 1,2,3 ($n_4 = 3$)

Then:

$$H = \begin{bmatrix} 0 & 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{2} & 0 \end{bmatrix}$$

Check columns: $\text{col}1 = 0 + 1/3 + 1/3 = 2/3$, etc. H describes a Markov chain — random surfer following random outlink.

If we tried to solve $H\pi = \pi$, we get stationary distribution of this Markov chain. But two problems:

1. **Dangling nodes:** Pages with no outlinks make column sum 0, not stochastic.
2. **Disconnected components / rank sinks:** Random surfer can get trapped in loop, never leave. Eigenvalue 1 may have multiplicity >1 or zero entries.

The Google Fix: Teleportation

PageRank assumes a "random surfer" who follows links with probability $d \approx 0.85$ and with probability $1 - d = 0.15$ gets bored and jumps to a uniformly random page anywhere on web.

Google matrix:

$$G = dH + \frac{1-d}{n}E$$

where E is $n \times n$ matrix of all ones, $n = 4$ pages, d damping factor usually 0.85. Second term $= (1 - d)/n$ everywhere.

Now G is positive (all entries >0), column-stochastic, irreducible and aperiodic. By Perron-Frobenius theorem for positive matrices:

- Spectral radius $\rho(G) = 1$ is simple eigenvalue
- Eigenvalue 1 has strictly positive eigenvector π unique up to scaling
- All other eigenvalues $|\lambda| < 1$, in fact $|\lambda_2| \leq d$

So power method converges quickly.

With $d = 0.85$, $n = 4$:

$$G = 0.85H + 0.0375 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$G \approx \begin{bmatrix} 0.0375 & 0.8875 & 0.4625 & 0.3208 \\ 0.3208 & 0.0375 & 0.0375 & 0.3208 \\ 0.3208 & 0.0375 & 0.0375 & 0.3208 \\ 0.3208 & 0.0375 & 0.4625 & 0.0375 \end{bmatrix}$$

The Eigenvector Equation

PageRank vector π satisfies

$$G\pi = \pi$$

This is eigenvector equation with eigenvalue $\lambda = 1$ (Bryan and Leise 572-575). Normalization $\sum_i \pi_i = 1, \pi_i \geq 0$ — it is probability distribution of where random surfer is after long time.

Solve $(G - I)\pi = 0$. For our 4-page example, solving gives (up to rounding):

$$\pi = \begin{bmatrix} 0.387 \\ 0.213 \\ 0.177 \\ 0.223 \end{bmatrix}$$

Interpretation:

- Page 1 receives 38.7% of importance — top result. Why? 3 pages link to it (2,3,4), including Page 2 which gives it all its importance.
- Page 4: 22.3%
- Page 2: 21.3%
- Page 3: 17.7% last.

Ranking: $1 > 4 > 2 > 3$. If you search query matching all 4, Page 1 returned first.

In real web of 1998: $n = 25$ million, H is $25M \times 25M$ but sparse — average 10 outlinks per page, so only ~250M nonzeros. Storing dense G impossible. Power method uses sparsity:

Initialize $\pi^{(0)} = (1/n)\mathbf{1}$, iterate:

$$\pi^{(k+1)} = G\pi^{(k)} = dH\pi^{(k)} + \frac{1-d}{n}\mathbf{1}$$

Because H sparse, $H\pi^{(k)}$ is $O(n)$. Convergence rate $O(d^k)$ — error after k steps $\approx d^k$. With $d = 0.85$, 50-100 iterations enough for tolerance 10^{-9} even for $n = 8$ billion today.

This is why PageRank scaled — it is power iteration for dominant eigenvector of sparse Markov chain.

Why Eigenvalue 1?

G is column-stochastic: columns sum to 1, because H columns sum to 1 and E columns sum to n times $(1-d)/n$. So $\mathbf{1}^T G = \mathbf{1}^T$. So 1 is eigenvalue with left eigenvector $\mathbf{1}^T$. Since spectrum of A and A^T same, G has right eigenvector with eigenvalue 1 — our π .

Perron-Frobenius guarantees $\pi > 0$ and unique, solving rank sink problem.

Intuition: Flow of Importance

Think of π as fluid. Each page j has π_j amount. It keeps $(1-d)\pi_j$ to distribute uniformly via teleportation, and sends $d\pi_j/n_j$ along each outlink. Equilibrium when inflow = outflow for each page — stationary distribution. Important pages are those

with large inflow — many inlinks from important pages.

This is eigenvector centrality — same math used for social influence, protein importance, etc. Google's insight was web's link graph encodes human judgment; eigenvector extracts global consensus from local votes.

Modern search: PageRank now one of hundreds of signals (BERT, freshness, location), but original principle — web as Markov chain, importance as dominant eigenvector of G — remains foundational architecture that turned a linear algebra problem into a \$2 trillion company.

Game Theory

Game theory is a mathematical framework for analyzing strategic interactions between rational decision-makers, where the outcome for each participant depends on the actions of others (Binmore 25-27). It models scenarios involving conflict or cooperation to identify optimal strategies, commonly used in economics, political science, evolutionary biology, and social sciences to predict behaviors (Resnik 121-125).

It was formalized by von Neumann and Morgenstern in *Theory of Games and Economic Behavior* (1944) and revolutionized by John Nash's equilibrium concept in 1950.

The notation of payoff matrices $u_i(s_1, s_2, \dots, s_n)$ and Nash equilibrium conditions formalizes thinking that humans already do implicitly in countless social situations — haggling, poker, driving in traffic, dating. While formal game theory requires mathematical sophistication, strategic competence — anticipating others' responses and optimizing accordingly — operates constantly in daily life, unrecognized as "mathematics" by those who claim to be "bad at math" (Stone 240-244; Rubinstein 140-146).

Formal Setup

A finite game in normal form is:

- $N = \{1, \dots, n\}$ players
- For each player i , a strategy set S_i — pure strategies available
- For each player i , a payoff / utility function $u_i : S_1 \times \dots \times S_n \rightarrow \mathbb{R}$

A strategy profile $s = (s_1, \dots, s_n) \in S$ determines payoff $u_i(s)$ to each player.

Assumptions: Players are rational — they maximize expected utility; they are intelligent — they know the game, know others are rational, know others know they know, etc. — common knowledge of rationality.

Game theory provides a rigorous mathematical framework, often requiring knowledge of calculus and real analysis, particularly for finding "existence proofs" of solutions such as Nash's theorem (Binmore 28-30). The core goal is to determine the best action for a player when outcome depends on actions of others (Resnik 125-128). Interestingly, game theory exhibits uncertainty principles analogous to quantum mechanics, where certain strategic information cannot be simultaneously optimized (Székely and Rizzo 688-695) — e.g., you cannot simultaneously maximize payoff and guarantee minimax in some games.

Payoff Matrices

Used to visualize and calculate results of simultaneous-move games for each player based on their choices (Resnik 130-135). Matrix entries represent utilities $u_i(s_1, s_2)$.

Classic 2-player, 2-strategy Prisoner's Dilemma. Two suspects, Cooperate = stay silent, Defect = betray. Payoffs = negative years in prison, higher is better:

	B: COOPERATE	B: DEFECT
A: Cooperate	(-1, -1)	(-3, 0)
A: Defect	(0, -3)	(-2, -2)

First number u_A , second u_B . If both cooperate, 1 year each. If A defects, B cooperates, A free, B gets 3 years. Both defect, 2 years each. What happens? See Nash below.

Nash Equilibrium

Nash Equilibrium: A situation where no player can benefit by unilaterally changing strategies while others keep theirs unchanged (Binmore 30-32). It is a stable fixed point of best-response reasoning — no one wants to deviate alone.

Formally, strategy profile $(s_1^*, s_2^*, \dots, s_n^*)$ is a Nash equilibrium if for every player i and every alternative strategy $s_i \in S_i$:

$$u_i(s_1^*, \dots, s_i^*, \dots, s_n^*) \geq u_i(s_1^*, \dots, s_i, \dots, s_n^*)$$

where u_i is player i 's utility function. It is foundational for predicting stable outcomes in competitive scenarios (Resnik 135-140).

In Prisoner's Dilemma, (Defect, Defect) is unique Nash equilibrium: if B defects, A's best response is defect (-2 > -3); if B cooperates, A's best response is still defect (0 > -1). Defect dominates. So rational play leads to (-2,-2), worse than (-1,-1) — dilemma.

Nash's Theorem (1950): Every finite game has at least one Nash equilibrium in mixed strategies. Proof uses Kakutani fixed point theorem — requires calculus and real analysis (Binmore 28-30). Existence proof is non-constructive.

Mixed Strategies and Utility Maximization

Often no pure Nash exists. Rock-Paper-Scissors has no pure equilibrium — any pure strategy is beaten.

Players assign numerical values (utility) to outcomes, acting to maximize own expected utility (Binmore 26-28). To guarantee equilibrium, allow randomization.

Mixed strategy σ_i is probability distribution over S_i . Let $\sigma_i(s_i)$ = prob player i plays s_i . Profile $\sigma = (\sigma_1, \dots, \sigma_n)$ induces distribution over outcomes:

Expected utility for mixed strategies is computed as

$$E[u_i] = \sum_{s \in S} p(s) \cdot u_i(s) = \sum_{s_1 \in S_1} \dots \sum_{s_n \in S_n} \left(\prod_{j=1}^n \sigma_j(s_j) \right) u_i(s_1, \dots, s_n)$$

where $p(s)$ is probability of strategy profile s — product if independent randomization.

Nash equilibrium in mixed: σ^* such that for all i , σ_i^* maximizes $E[u_i]$ given σ_{-i}^* .

Rock-Paper-Scissors equilibrium: each player plays (1/3,1/3). Expected payoff 0. Any deviation exploited.

Indifference principle: In mixed equilibrium, player randomizes to make opponent indifferent among strategies in support. This gives linear equations to solve.

Mathematical Techniques

The mathematics needed scales with the game. Simple parlor games need arithmetic; general existence and computation need topology, convex analysis, and fixed-point theorems (Weil 360-363; Binmore 25-34).

1. Basic 2x2 — Arithmetic, Dominance, Best-Response

For finite games with 2 players, 2 strategies each, you can solve by hand.

- **Strict dominance:** Strategy s_i strictly dominates s'_i if $u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$ for all s_{-i} . Rational player never plays dominated strategy — eliminate it. Iterate: iterated elimination of strictly dominated strategies (IESDS) yields unique prediction in Prisoner's Dilemma, many games.
- **Best-response:** For each column (opponent's strategy), mark row player's highest payoff; for each row, mark column player's highest payoff. Cells where both payoffs marked are pure Nash equilibria.
- Example: Stag Hunt

	HARE	STAG
Hare	2,2	2,0
Stag	0,2	3,3

Best responses: If opponent plays Hare, you prefer Hare ($2 > 0$). If opponent plays Stag, you prefer Stag ($3 > 2$). Two pure Nash: (Hare,Hare) and (Stag,Stag) — coordination problem.

2. Calculus — Continuous Strategies

For continuous $S_i = [0, \infty)$ like Cournot duopoly where firms choose quantity q_i , market price $P(Q) = a - bQ$, $Q = q_1 + q_2$, cost cq_i , profit:

$$u_i(q_1, q_2) = q_i(a - b(q_1 + q_2)) - cq_i$$

Nash equilibrium solves first-order condition $\partial u_i / \partial q_i = 0$ with $\partial^2 u_i / \partial q_i^2 < 0$ for max:

$$\partial u_1 / \partial q_1 = a - b(2q_1 + q_2) - c = 0 \implies q_1 = \frac{a - c - bq_2}{2b} = BR_1(q_2)$$

Similarly $BR_2(q_1) = \frac{a - c - bq_1}{2b}$. Intersection of best-response functions gives Nash:

$$q_1^* = q_2^* = \frac{a - c}{3b}$$

This method — best-response calculus — extends to any differentiable u_i : solve $\nabla_{s_i} u_i = 0$. Requires checking second-order and boundary.

For Hotelling location, Bertrand pricing, public goods contribution games, calculus yields interior equilibria; often need Kuhn-Tucker for inequality constraints.

3. Linear Algebra — Zero-Sum, Lemke-Howson, Perron

Zero-sum game $u_1 = -u_2 = A$ payoff matrix for row player.

Row player chooses mixed $\mathbf{x} \geq 0, \sum x_i = 1$ to maximize $\min_j \mathbf{x}^T A e_j$. Column player chooses $\mathbf{y}$ to minimize $\max_i e_i^T A \mathbf{y}$.

Von Neumann's Minimax theorem:

$$\max_{\mathbf{x}} \min_{\mathbf{y}} \mathbf{x}^T A \mathbf{y} = \min_{\mathbf{y}} \max_{\mathbf{x}} \mathbf{x}^T A \mathbf{y} = v^*$$

Value v^* found via linear programming duality:

Primal: $\max v$ s.t. $A^T \mathbf{x} \geq v \mathbf{1}, \sum x_i = 1, \mathbf{x} \geq 0$ Dual: $\min v$ s.t. $A \mathbf{y} \leq v \mathbf{1}, \sum y_j = 1, \mathbf{y} \geq 0$

Solving LP via simplex gives equilibrium. Eigenvector methods appear when A is symmetric — equilibrium proportional to Perron eigenvector.

For general bimatrix games (A, B) not zero-sum, Lemke-Howson algorithm follows a path of almost-completely-labeled polytopes defined by linear inequalities $A \mathbf{y} \leq \mathbf{1}, B^T \mathbf{x} \leq \mathbf{1}$ to find Nash. It is pivoting like simplex. Complexity is PPAD-complete — believed hard, but practical for moderate size.

4. Probability — Mixed Strategies and Minimax

For mixed strategies, expected utility is linear in probabilities:

$$E[u_i](\sigma) = \sum_s \left(\prod_j \sigma_j(s_j) \right) u_i(s)$$

Linearity yields key property:

Indifference principle: If σ^* is Nash and player i mixes between s_i and s'_i with positive probability, then $E[u_i(s_i, \sigma^*)] = E[u_i(s'_i, \sigma^*)]$ — otherwise he'd shift to higher.

So to find mixed equilibrium, set opponent indifferent.

Example: Matching Pennies $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Let column mix $(q, 1 - q)$. Row payoff for playing Top: $q * 1 + (1 - q) * (-1) = 2q - 1$. Bottom: $q * (-1) + (1 - q) * 1 = 1 - 2q$. Indifference $2q - 1 = 1 - 2q \implies q = 1/2$. So equilibrium $(1/2, 1/2)$.

Minimax theorem: $\max_{\sigma_1} \min_{\sigma_2} E[u] = \min_{\sigma_2} \max_{\sigma_1} E[u]$ — order of pessimism doesn't matter in zero-sum. Proof uses separating hyperplane.

5. Fixed Point Topology — Existence Proof

Nash's 1950 existence theorem: Every finite game has at least one mixed Nash equilibrium.

Proof sketch: Define best-response correspondence $BR(\sigma) = \{\tau : \tau_i \in \arg \max E[u_i(\cdot, \sigma_{-i})]\}$. Show BR is upper hemicontinuous, convex-valued, maps compact convex simplex $\Delta = \prod \Delta_i$ to itself. Apply Kakutani fixed-point theorem — a generalization of Brouwer fixed-point — there exists $\sigma^* \in BR(\sigma^*)$, which is Nash. Requires real analysis, topology. Same technique gives Arrow-Debreu existence of competitive equilibrium in economics.

6. Dynamic Programming / Backward Induction — Extensive Form

For sequential games, represent as game tree with information sets. Use backward induction to find subgame perfect equilibrium (SPE) — Nash in every subgame, eliminating non-credible threats.

Algorithm: Start at leaves, compute payoffs, move up: at decision node of player i , choose action maximizing u_i given already-solved continuation. Replace node with that payoff.

Example: Entry game: Entrant chooses In/Out, then Incumbent chooses Fight/Accommodate. Payoffs: (Out:0,2), (In,Fight:-1,-1), (In,Accommodate:1,1). Backward induction: If Entrant In, Incumbent prefers Accommodate ($1 > -1$). So Entrant anticipates $1 > 0$, chooses In. SPE (In, Accommodate), even though (Out, Fight) is Nash with threat "I will fight" — not credible, because Fight not optimal off-path.

For imperfect information, need sequential rationality and beliefs — Perfect Bayesian Equilibrium solved via Bayes rule + backward induction.

Types and Extensions

1. Cooperative vs Non-cooperative

- **Non-cooperative:** Studies individual strategic choice, self-enforcing agreements. Solution: Nash equilibrium. Focus of most modern theory.
- **Cooperative:** Allows binding agreements, coalitions. Question: how to divide surplus? N players, characteristic function $v(C)$ value coalition C can achieve. Solutions:
 - **Core:** allocations where no coalition can deviate and do better for all its members — $\sum_{i \in C} x_i \geq v(C)$ for all C , $\sum_{i \in N} x_i = v(N)$.
 - **Shapley value:** Fair division based on marginal contribution averaged over all arrival orders: $\phi_i(v) = \frac{1}{n!} \sum_{\pi} [v(P_i^{\pi} \cup \{i\}) - v(P_i^{\pi})]$, where P_i^{π} players before i in permutation π . Axiomatizes fairness.

2. Zero-Sum vs Non-Zero-Sum

- **Zero-sum:** $u_1 + u_2 = 0$ — strictly competitive, one's gain is other's loss. Poker simplified, chess. Minimax, value. No cooperation possible.
- **Non-zero-sum:** General $u_1 + u_2 \neq 0$ — allows both conflict and cooperation. Prisoner's Dilemma, Stag Hunt, Chicken. Win-win possible, but also dilemma. Nash equilibria may be Pareto inefficient — (Defect,Defect) Pareto dominated by (Cooperate,Cooperate).

3. Repeated Games

Same stage game G played infinitely ($t = 0, 1, \dots$) with discount factor $\delta \in (0, 1)$ — future payoff weighted δ^t . Total payoff $\sum \delta^t u_i(s^t)$.

Folk theorem: If players patient enough δ close to 1, any feasible payoff above minmax can be sustained as subgame perfect equilibrium via threat strategies — e.g., Grim Trigger: cooperate until someone defects, then punish forever. So cooperation can be sustained by threats even when stage Nash is defect — explains social norms, cartels, relational contracts. Tit-for-Tat (cooperate initially, copy opponent's last move) famously robust in Axelrod tournaments.

Mathematically, uses one-shot deviation principle and dynamic programming.

4. Evolutionary Game Theory

Strategies are phenotypes encoded in genes, payoff is fitness (expected offspring). Population state $\mathbf{x}$ frequencies. Replicator dynamics: $\dot{x}_i = x_i(u_i(\mathbf{x}) - \bar{u}(\mathbf{x}))$ — strategies with above-average fitness grow.

Equilibrium concept: Evolutionarily Stable Strategy (ESS) — strategy σ resistant to invasion by small mutant fraction ϵ : $u(\sigma, (1 - \epsilon)\sigma + \epsilon\tau) > u(\tau, (1 - \epsilon)\sigma + \epsilon\tau)$ for all $\tau \neq \sigma$ near. Stronger than Nash: Nash is resistant to best-response, ESS resistant to mutant that could do equally well. Example: Hawk-Dove game: V resource, C cost of fight. Pure Hawk not ESS if $C > V$, mixed ESS with $p = V/C$ probability Hawk.

Explains animal conflict, cooperation without rationality — just natural selection.

5. Bayesian Games — Incomplete Information

Players have private types t_i — e.g., valuation in auction, cost in oligopoly, ability. Types drawn from common prior $p(t)$. Player knows own type, not others'. Strategy is type-dependent $s_i(t_i)$.

Payoff $u_i(s_1, \dots, s_n; t_1, \dots, t_n)$. Equilibrium: Bayes-Nash equilibrium where each type t_i maximizes expected utility given beliefs about others' types conditional on own type:

$$\sigma_i^*(\cdot|t_i) \in \arg \max \mathbb{E}_{t_{-i}|t_i}[u_i(\sigma_i, \sigma_{-i}^*(t_{-i}); t_i, t_{-i})]$$

Foundation of auction theory (Vickrey, Myerson), signaling games (Spence education), screening, mechanism design — designing games to achieve desired outcome — e.g., second-price auction is dominant-strategy incentive compatible, revenue equivalence theorem.

In all cases, mathematics turns "what will others do?" into fixed-point, optimization, or dynamical systems problems — making intuition precise and testable.

Game theory is the mathematics of $u_i(s_i, s_{-i})$ — your payoff depends on others' choices. Nash equilibrium is the point where expectations are consistent and no one regrets their choice given others' choices — the natural rest point of strategic reasoning.

Prisoner's Dilemma

Two suspects are arrested and interrogated separately, no communication. Prosecutor offers deal. Each has two strategies:

- **Cooperate** (with each other, stay silent — C)
- **Defect** (betray the other, confess — D)

Note: terminology is confusing — Cooperate means cooperate with accomplice, defect from prosecutor, and vice versa. Game theorists often use C/D.

The payoff matrix shows years in prison as negative utility — higher numbers better, less prison. Entry $(a, b) = (\text{Suspect 1's years}, \text{Suspect 2's years})$:

	SUSPECT 2: COOPERATE	SUSPECT 2: DEFECT
Suspect 1: Cooperate	(-1, -1)	(-10, 0)
Suspect 1: Defect	(0, -10)	(-5, -5)

Interpretation: If both silent, police have minor evidence: 1 year each. If one confesses and other silent, confessor goes free 0, silent gets 10. If both confess, 5 years each — reduced from 10 for cooperation with police, but worse than mutual silence.

Dominance Reasoning

From Suspect 1's perspective:

- If Suspect 2 cooperates: Defect gives 0 years vs. Cooperate gives 1 year → **Defect is better** by 1 year.
- If Suspect 2 defects: Defect gives 5 years vs. Cooperate gives 10 years → **Defect is better** by 5 years.

Defecting yields strictly higher payoff regardless of opponent's choice. It strictly dominates Cooperate.

Defecting is a **dominant strategy** — optimal regardless of opponent's choice (Cunningham 15-18). Dominant strategy equilibrium concept is stronger than Nash: it doesn't require belief about opponent.

By symmetry, Suspect 2 has identical reasoning. Both rationally choose to defect, yielding outcome **(-5, -5)**.

Nash Equilibrium

(Defect, Defect) is the unique Nash equilibrium (Cunningham 18-20). At this point, no unilateral deviation helps:

$$u_1(\text{Defect, Defect}) = -5 \geq u_1(\text{Cooperate, Defect}) = -10$$

$$u_2(\text{Defect, Defect}) = -5 \geq u_2(\text{Defect, Cooperate}) = -10$$

Check deviations: If 1 deviates to Cooperate while 2 stays Defect, 1 goes from -5 to -10 — worse. So Nash.

Mutual cooperation (-1,-1) would be better for both than mutual defection (-5,-5) — 4 years less each. But it is not stable: each has temptation to deviate to 0. If both at (-1,-1), each thinks "if I defect while other cooperates, I go free."

This illustrates how individual rationality can lead to collective irrationality — a fundamental insight with implications for environmental policy, arms races, and public goods provision (Rubinstein 100-110). (Cooperate, Cooperate) Pareto dominates (Defect,Defect) — both better — but not Nash. (Defect,Defect) is Nash but Pareto inefficient. That tension is the dilemma.

General definition: Prisoner's Dilemma requires payoffs $T > R > P > S$ where T =Temptation to defect, R =Reward mutual cooperate, P =Punishment mutual defect, S =Sucker's payoff. Here $0 > -1 > -5 > -10$.

Why It Matters Everywhere

- **Environmental:** Two firms can pollute (Defect) cheaply or abate (Cooperate) costly. Mutual abatement good for both long-term, but each has incentive to free-ride and pollute while other abates.
- **Arms race:** Cooperate = disarm, Defect = arm. Mutual disarm best, but each fears being exploited, so both arm.
- **Public goods:** Cooperate = contribute, Defect = free ride. Doping in sports, overfishing, climate change — all PD.
- **Pricing:** Two firms can keep prices high (Cooperate) or undercut (Defect). Mutual high price profitable, but each wants to undercut to steal market — leads to price war.

Repeated Prisoner's Dilemma — How Cooperation Emerges

When game repeats indefinitely, cooperation can emerge through conditional strategies (Resnik 155-165).

One-shot PD predicts defection. Real humans cooperate often. Why? Because life is repeated. If you defect today, opponent can punish tomorrow.

Let game repeated with discount factor $\delta \in (0, 1)$ — next period's payoff worth δ times today's. Patience high = δ close to 1.

Strategies:

- **Grim Trigger:** Cooperate until opponent defects once, then defect forever. Cooperation sustainable if δ large enough to make one-time temptation T not worth future loss. Condition: $R/(1 - \delta) \geq T + \delta P/(1 - \delta) \rightarrow \delta \geq (T - R)/(T - P)$. With our numbers $R = -1, P = -5, T = 0$: Need $\delta \geq (0 - (-1))/(0 - (-5)) = 1/5 = 0.2$. So if future matters at least 20% as much as present, cooperation possible.
- **Tit-for-Tat:** Start cooperating, then mirror opponent's previous move. Anatol Rapoport's strategy won Axelrod's 1980 tournament — nice, retaliatory, forgiving, clear. It rewards cooperation, punishes defection, but forgives.
- **Tit-for-Two-Tats, Generous Tit-for-Tat, Win-Stay-Lose-Shift:** Variants that handle noise — if opponent defects by mistake, forgiving versions avoid endless retaliation spiral.

Folk theorem: With sufficient patience (low discount rate, high δ), nearly any feasible payoff between full defection and full cooperation can be sustained as subgame perfect equilibrium in infinitely repeated game. Formally, for any feasible individually rational payoff profile v with $v_i > P$, exists $\bar{\delta} < 1$ such that for all $\delta > \bar{\delta}$, v is SPE.

Implications: Repeated interaction enables social norms, trust, reputation. Institutions that make future interaction likely (communities, long-term contracts) promote cooperation. One-shot anonymous interaction — e.g., one-time online transaction — predicts defection, hence need for enforcement.

Helping a Coworker — Prisoner's Dilemma at Work

You and a coworker are both working on a project that affects both your reputations — presentation to boss, shared OKR. Each can choose:

- **Help** — contribute extra effort, stay late, polish deck
- **Slack** — minimal effort, free-ride

Payoffs represent net benefit = recognition minus effort cost:

	COWORKER: HELP	COWORKER: SLACK
You: Help	(3, 3)	(-1, 4)
You: Slack	(4, -1)	(0, 0)

Interpretation:

- **(Help, Help) = (3, 3):** Both contribute, project succeeds impressively, both get credit minus effort cost. 5 units credit -2 effort =3. Team win.
- **(Help, Slack) = (-1, 4):** You work hard while they coast; they get credit, you're exhausted. You: 1 credit -2 effort = -1; they: 4 credit -0 effort =4. Sucker's payoff.
- **(Slack, Help) = (4, -1):** You coast while they work; you get credit without effort — temptation.
- **(Slack, Slack) = (0, 0):** Project mediocre, no one looks good, minimal effort wasted. Baseline.

This is Prisoner's Dilemma with $T = 4 > R = 3 > P = 0 > S = -1$.

Analysis

From your perspective:

- If coworker helps: Slacking gives 4 vs. Helping gives 3 → **Slack is better** by +1 gain — free credit.
- If coworker slacks: Slacking gives 0 vs. Helping gives -1 → **Slack is better** by +1 gain — avoid being sucker.

Slacking strictly dominates Helping. No matter what coworker does, you are +1 better by slacking.

Nash Equilibrium: (Slack, Slack) with payoff **(0,0)** (Binmore 30-32). Neither can improve unilaterally:

$$u_{\text{you}}(\text{Slack}, \text{Slack}) = 0 \geq u_{\text{you}}(\text{Help}, \text{Slack}) = -1$$

$$u_{\text{coworker}}(\text{Slack}, \text{Slack}) = 0 \geq u_{\text{coworker}}(\text{Slack}, \text{Help}) = -1$$

Check: If you are at (0,0) and you deviate to Help, you go to -1 — worse. So Nash.

But mutual helping **(3,3)** is Pareto superior — better for everyone, total welfare 6 vs 0 (Cunningham 24-26). No one wants to move to (3,3) alone because deviator gains 1, but both moving helps both. This creates workplace tension: rational self-interest suggests slacking, but everyone worse off than if they'd cooperated. Classic social dilemma.

Why workplaces feel draining? Because many tasks are PD: documentation, cleaning shared kitchen, mentoring. Each person prefers others do it.

Real-World Modifications — How Cooperation Emerges

The 2×2 one-shot model misses structure that saves workplaces.

1. Repeated interaction — Shadow of the Future

If you work together repeatedly — same team for months — defecting (slacking) now damages future cooperation. Game becomes repeated Prisoner's Dilemma (Resnik 155-160).

Let discount factor δ = how much you value next week's interaction vs this week's. With Grim Trigger — "I will Help as long as you Help; if you Slack once, I Slack forever" — cooperation sustained if:

$$R + \delta R + \delta^2 R + \dots \geq T + \delta P + \dots$$

$$\frac{3}{1-\delta} \geq 4 + \frac{0}{1-\delta} \implies 3 \geq 4(1-\delta) \implies \delta \geq 0.25$$

If you care at least 25% about future, cooperation rational. In long-term teams δ high, cooperation emerges naturally. In one-off project with contractor you never see again $\delta \approx 0$, slack expected.

Practical strategies people use without naming them:

- **Tit-for-Tat:** Start Help, then copy coworker's last move. "I'll help if they help, but I'm not getting taken advantage of" — exactly computing conditional strategies in implicit repeated game (van Benthem et al. 126-129). Nice, retaliatory, forgiving, clear.
- **Win-Stay-Lose-Shift:** If last outcome (3,3) or (4,-1) good for you, repeat; if (-1,4) or (0,0) bad, switch. Helps recover from errors.

2. Reputation — Beyond Dyad

In office environments, your choice affects reputation beyond one coworker. Single-shot payoffs don't capture long-term career costs of being known as slacker (Rubinstein 110-120).

Model: Add third party observers. If you Slack, others update belief about you, reduce future opportunities. Effective payoff of Slack when coworker Helps becomes $4 - r$, where r = reputation cost. If $r > 1$, Help becomes dominant. Companies create this intentionally via 360 reviews, public kudos, visibility.

So (Help,Help) can be sustained by indirect reciprocity: "I help you because others watch and will help me later."

3. Altruism / Reciprocity / Inequality Aversion

People often have utility functions that value fairness and reciprocity beyond pure self-interest, changing effective payoff matrix (Rubinstein 120-130).

- **Altruism:** $u_i^{\text{eff}} = u_i + \alpha u_j$, $\alpha > 0$ care about coworker. If $\alpha > 0.33$, effective payoff of (Help,Help) becomes $3 + 3\alpha > 4 - \alpha$ of (Slack,Help), so Help dominates.
- **Inequality aversion:** Dislike unfair outcomes: $u_i^{\text{eff}} = u_i - \beta \max(u_j - u_i, 0)$. (Slack,Help) gives inequality 5 for sucker, so psychological cost high.
- **Reciprocity:** Utility depends on intentions — being kind to kind person feels good.

Experimentally, in lab PD with payoffs (3,3)/(0,4) etc., ~50% cooperate one-shot, not 0% — because social preferences transform matrix.

4. Mechanism Design — How Managers Fix PD

Managers can change game to make (Help,Help) equilibrium:

- **Make contributions observable:** Daily standup, shared commits — increase r , turns hidden Slack into visible.
- **Joint accountability + individual accountability:** Split credit by observable contribution, not equal split. Change payoffs from (4,-1) to (2,-1) if slacker gets less credit.
- **Reduce effort cost:** Tools, templates — increase Help payoff from 3 to 4.5.
- **Increase P penalty for mutual slacking:** If project fails, both get -2 not 0, (Slack,Slack) worse than being sucker, transforms game into Stag Hunt with two equilibria, coordination on Help possible.
- **Formal contracts:** If you can contract "if you Slack, you owe me", enforceable commitment changes game.

In one line: Workplace Help game is PD with $T > R > P > S$. One-shot logic says Slack, but repetition, reputation, and social preferences turn it into a repeated game where Tit-for-Tat and observable contribution make Help rational — which is why teams with long horizon and transparency outperform collections of rational slackers.

Fourier Transform

The continuous Fourier Transform is the ultimate prism: it decomposes a signal into its pure frequencies.

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

where $f(t)$ is time-domain signal, $F(\omega)$ is frequency-domain representation — complex amplitude of frequency ω present in f (Berry 227-230). $e^{-i\omega t} = \cos \omega t - i \sin \omega t$ by Euler, so integral correlates f with sine/cosine at ω .

The inverse transform reconstructs original signal from frequencies:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega$$

This bidirectional relationship — Fourier Transform Identity Theorem — guarantees information perfectly preserved in both representations: $f \leftrightarrow F$ is bijection (in $L^1 \cap L^2$, extended to L^2 by Plancherel). Time domain tells you *when*, frequency domain tells you *what pitch* (Berry 230-232, 227). Both are same signal.

Intuition: Eigenfunction Expansion

$e^{i\omega t}$ are eigenfunctions of differentiation d/dt with eigenvalue $i\omega$: $\frac{d}{dt}e^{i\omega t} = i\omega e^{i\omega t}$. They are also eigenfunctions of translation and convolution operators. Fourier transform is change of basis to eigenbasis of time-shift invariant linear systems — diagonalizing them. Same story as eigenvectors: F is coordinates of f in eigenbasis $\{e^{i\omega t}\}$.

Properties — Why Engineers Live In Frequency Domain

Let $F(\omega) = \mathcal{F}\{f\}$, $G = \mathcal{F}\{g\}$.

- **Linearity:** $\mathcal{F}\{af + bg\} = aF + bG$
- **Time shift:** $\mathcal{F}\{f(t - a)\} = e^{-i\omega a}F(\omega)$ — delay = phase rotation
- **Frequency shift:** $\mathcal{F}\{e^{i\omega_0 t}f(t)\} = F(\omega - \omega_0)$ — modulation
- **Scaling:** $\mathcal{F}\{f(at)\} = \frac{1}{|a|}F(\omega/a)$ — compress in time, spread in frequency
- **Differentiation:** $\mathcal{F}\{f'(t)\} = i\omega F(\omega)$ — ODE becomes algebra, like Laplace but for all $t \in \mathbb{R}$
- **Convolution:** $\mathcal{F}\{f * g\} = F(\omega)G(\omega)$, where $(f * g)(t) = \int f(\tau)g(t - \tau)d\tau$. And $\mathcal{F}\{fg\} = \frac{1}{2\pi}F * G$ dual. Filtering is multiplication in frequency.
- **Parseval / Plancherel:** Energy preserved: $\int |f(t)|^2 dt = \frac{1}{2\pi} \int |F(\omega)|^2 d\omega$. Power spectrum $|F|^2$ tells where energy lies.

Convergence and Uncertainty

Mean convergence theorems ensure Fourier representations converge to original function under broad conditions — e.g., if $f \in L^2$, partial integrals $\int_{-R}^R F(\omega)e^{i\omega t}d\omega$ converge in L^2 mean to f as $R \rightarrow \infty$ (McShane 205-208). Pointwise convergence needs more: Dirichlet conditions piecewise smooth $\rightarrow$ convergence to midpoint of jump.

The Fourier Transform satisfies remarkable inequalities that constrain how "spread out" a function can be simultaneously in time and frequency domains (Beckner 159-165). These are uncertainty principles, formalized through weighted norm inequalities, with profound implications from quantum mechanics to signal processing (Beckner 175-180; Muckenhoupt 729-735).

- **Heisenberg uncertainty:** Cannot be localized in both time and frequency. For f normalized, define variances $\Delta_t^2, \Delta_\omega^2$. Then $\Delta_t \Delta_\omega \geq 1/2$. Proof uses Cauchy-Schwarz and $\mathcal{F}\{tf(t)\} = iF'(\omega)$. Equality for Gaussian $e^{-t^2/2}$. In quantum mechanics $p = \hbar\omega$, this is $\Delta x \Delta p \geq \hbar/2$.
- **Hardy, Hausdorff-Young, Beckner:** Quantitative bounds: If $|f(t)| \leq Ce^{-at^2}$ and $|F(\omega)| \leq Ce^{-b\omega^2}$ with $ab > 1/4$, then $f = 0$. Weighted L^p norms $\|F\|_q \leq C\|f\|_p$ for $1/p + 1/q = 1$, $1 \leq p \leq 2$ — Beckner's sharp constant.
- **Implication:** No signal is both time-limited and band-limited — if $f(t) = 0$ outside interval, $F(\omega)$ cannot be 0 outside interval unless $f = 0$. This limits perfect filtering.

Generalizations and Modern Extensions

The transform extends beyond real and complex numbers to quaternions and higher algebraic structures, enabling analysis of multidimensional rotations and color image processing (Gao 9851-9860).

- **Discrete Time Fourier Transform (DTFT):** f discrete, $F(\omega) = \sum f[n]e^{-i\omega n}$, periodic in ω . [n]
- **Discrete Fourier Transform (DFT):** Both discrete, length N : $F[k] = \sum_{n=0}^{N-1} f[n]e^{-i2\pi kn/N}$. Computable via FFT $O(N \log N)$ — basis of digital signal processing.
- **Fourier Series:** Periodic $f(t + T) = f(t) \rightarrow$ discrete frequencies $F_n = \frac{1}{T} \int_0^T f(t)e^{-in\omega_0 t} dt$, $f = \sum F_n e^{in\omega_0 t}$ — our guitar string eigenfunctions.
- **n -D Fourier:** $F(\omega_1, \omega_2) = \iint f(x, y)e^{-i(\omega_1 x + \omega_2 y)} dx dy$ — image processing, low-pass blur = keep low ω , high-pass edge detection = keep high ω .
- **Quaternion Fourier Transform:** For $f : \mathbb{R}^2 \rightarrow \mathbb{H}$ quaternion-valued, allows processing color (RGB as i,j,k) while preserving cross-channel correlation — used in color image filtering, flow analysis. Noncommutative: left vs right transform differ.
- **Short-Time Fourier Transform (STFT) / Gabor:** To get time *and* frequency, window $f(t)g(t - \tau)$ then Fourier in t : spectrogram $|STFT(\tau, \omega)|^2$ — what you see in music apps. Tradeoff governed by uncertainty.
- **Fractional Fourier, Wavelet:** Intermediate bases.

Examples and Applications

Input Signals — Time Domain Hides What Frequency Domain Shows

Pure tone: Input signal $f(t) = \cos(\omega_0 t)$: mathematically simple, infinitely long cosine. Its Fourier Transform:

$$F(\omega) = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

— two spikes at $\pm\omega_0$ (Dirac deltas). Negative frequency appears because $\cos \omega_0 t = (e^{i\omega_0 t} + e^{-i\omega_0 t})/2$ — needs both. Real signals have Hermitian symmetry $F(-\omega) = F^*(\omega)$.

Noisy signal: Real world $y(t) = f(t) + n(t)$ where f desired signal band-limited to $|\omega| < \omega_c$, $n(t)$ high-frequency noise. In time domain $y(t)$ looks messy — signal buried. In frequency domain:

$$Y(\omega) = F(\omega) + N(\omega)$$

F concentrated near $\pm\omega_0$, N spread at high $|\omega|$. They separate. Filtering: multiply $Y(\omega)$ by ideal low-pass $H(\omega) = 1$ for $|\omega| < \omega_{cut}$, 0 otherwise, then inverse transform $y_{filtered}(t) = \frac{1}{2\pi} \int H(\omega)Y(\omega)e^{i\omega t} d\omega$ to recover clean signal. This is exactly how:

- **Noise cancellation:** Mic captures $S(t) + Noise(t)$, adaptive filter estimates *Noise* spectrum, subtracts in frequency domain.
- **MRI reconstruction:** MRI scanner samples k-space which *is* Fourier domain: k_x, k_y correspond to spatial frequencies. Inverse 2D Fourier $\mathcal{F}^{-1}\{K(k_x, k_y)\} = \text{image } I(x, y)$. Undersample k-space = blurry image.
- **MP3/AAC compression:** Human ear masks quiet frequencies near loud ones — psychoacoustic model keeps perceptually important $F(\omega)$ bins, discards 90% of coefficients. File size drops 10x, sounds same because discarded information inaudible. JPEG does same in 2D with DCT, a real version of Fourier.

In one line: Fourier transform is rotation of basis to eigenvectors of translation — turning convolution into multiplication, differentiation into scaling, and revealing hidden harmonic content that time domain obscures, at cost of Heisenberg tradeoff that you cannot be sharp in both domains.

Piano Chords — Hearing Eigenmodes

When you play a single note, say middle A at 440 Hz (A4), ideal sound wave — ignoring decay — can be approximated as:

$$A(t) = \sin(2\pi \cdot 440 \cdot t)$$

Pure sine sounds artificial, like tuning fork or synthesizer with no harmonics — not like piano.

When you play a chord — say A-major triad with notes A4 (440 Hz), C#5 (554.37 Hz), and E5 (659.25 Hz) — time domain waveform looks like messy wobble:

$$S(t) = A_1 \sin(2\pi \cdot 440t) + A_2 \sin(2\pi \cdot 554t) + A_3 \sin(2\pi \cdot 659t)$$

where A_1, A_2, A_3 amplitudes (loudness) of each note, dependent on velocity of keypress. Graph $S(t)$ vs t shows beating, interference — you cannot read frequencies by eye.

Fourier Transform decomposes composite wave:

$$\mathcal{F}\{S(t)\} = F(\omega)$$

producing frequency spectrum with three distinct peaks at 440 Hz, 554 Hz, and 659 Hz, with heights proportional to A_1, A_2, A_3 (Alm and Walker 461-465). Technically, $\mathcal{F}\{\sin \omega_0 t\} = i\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$, imaginary spikes. Magnitude $|F(\omega)|$ shows peaks.

This decomposition reveals exactly which notes present and relative volumes — information that exists in composite waveform but invisible in time domain. This is what Shazam does: computes spectrogram of audio, peaks correspond to chord notes, matches fingerprint to database.

Real Instruments: Harmonic Series Defines Timbre

Real instruments far more complex. A piano string doesn't produce pure sine wave; fixed ends quantize allowed modes, as in Eigenvalue section. It generates overtones — harmonics — at integer multiples of fundamental frequency: $f_n = n f_1$: 440 Hz, 880 Hz, 1320 Hz, 1760 Hz, etc. (Alm and Walker 465-470).

Fourier series of periodic piano A:

$$\text{Piano A}(t) = \sum_{n=1}^{\infty} a_n \sin(2\pi \cdot n \cdot 440 \cdot t + \phi_n)$$

where coefficients a_n decrease with n , roughly $a_n \sim 1/n$ for plucked/string with some envelope, and ϕ_n phases. For ideal string plucked at center, even $a_n = 0$.

The unique pattern of these overtones — relative strengths a_n — defines instrument's timbre (Alm and Walker 471-473).

- **Piano:** a_n decays slowly, many harmonics, plus inharmonicity due to string stiffness: actual partials slightly sharp $f_n = n f_1 \sqrt{1 + B n^2}$, $B \sim 10^{-4}$, gives piano its bright, slightly out-of-tune shimmer.
- **Violin:** Sawtooth-like bowing excites strong higher harmonics, $a_n \sim 1/n$, very rich — why violin sounds bright, cutting. Body resonances filter a_n further, formants.
- **Flute:** Near pure sine, a_1 dominant, $a_{n>1}$ tiny — hollow, pure.
- **Clarinet:** Closed-open pipe, only odd harmonics $n = 1, 3, 5, \dots$ — hence hollow quality.

A violin playing same note A4 has different pattern a_n values, which is why it sounds distinct from piano despite same fundamental f_1 . Your ear does Fourier transform mechanically: cochlea is frequency analyzer — basilar membrane different positions resonate at different ω , hair cells send $|F(\omega)|$ to brain. You *hear* Fourier transform.

Chord with harmonics: A-major chord on real piano:

$$S_{\text{real}}(t) = \sum_{n=1}^{\infty} a_n^{(A)} \sin(2\pi n 440t) + \sum_{m=1}^{\infty} a_m^{(C\#)} \sin(2\pi m 554t) + \sum_{k=1}^{\infty} a_k^{(E)} \sin(2\pi k 659t)$$

Spectrum $|F(\omega)|$ now shows clusters: peaks at 440, 554, 659 and each has harmonic comb at multiples. Some harmonics overlap nearly — e.g., 3rd harmonic of A = 1320 Hz near 2nd harmonic of E = 1318.5 Hz — they beat, creating richness and consonance/dissonance perception. Consonant chords have many overlapping harmonic partials aligning; dissonant chords have partials clashing within critical band ~15% of frequency.

Thus playing piano is physically adding eigenvectors of wave operator, and listening is computing their Fourier coefficients — your auditory system performing spectral decomposition to infer which keys pressed and which instrument played.